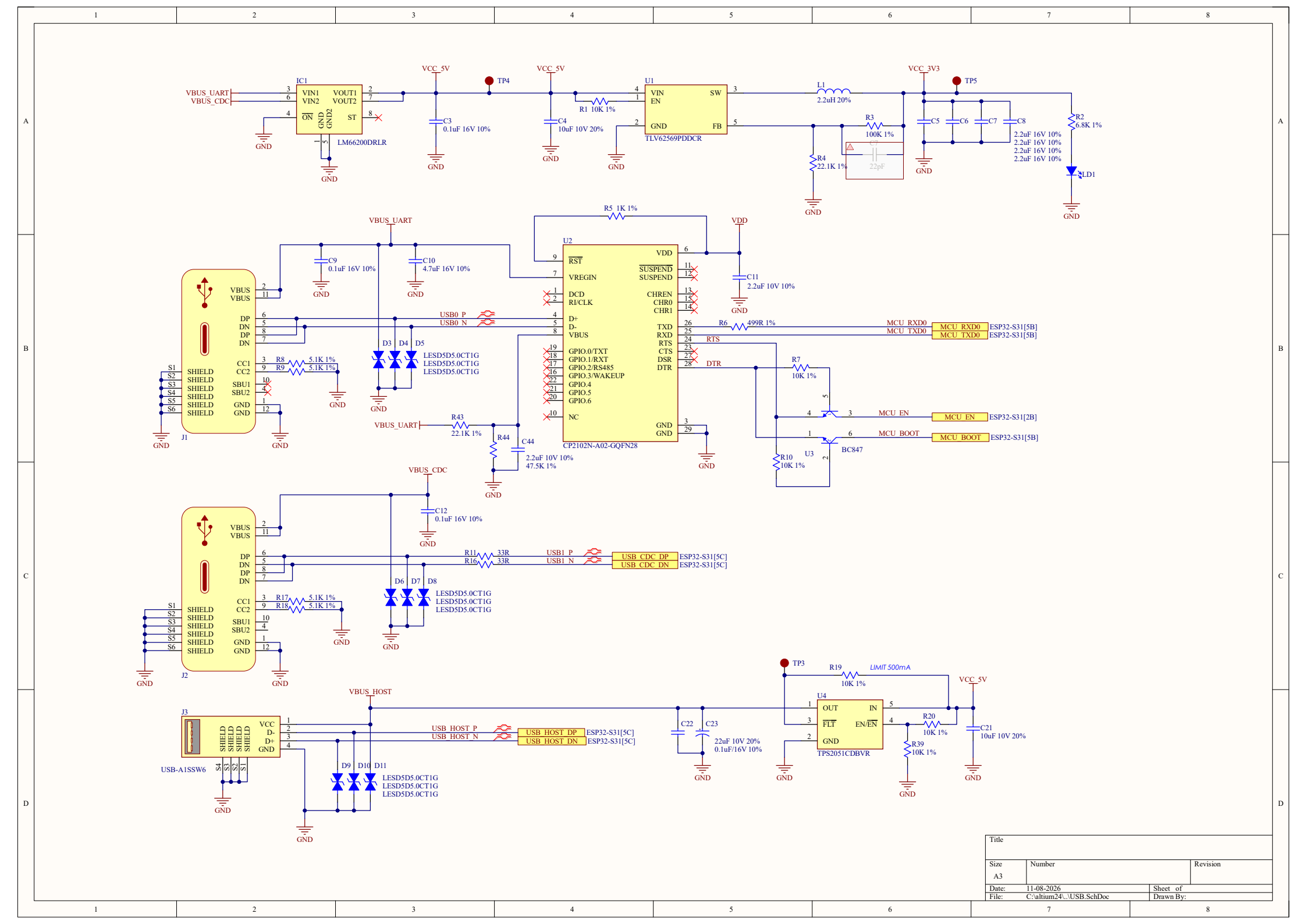
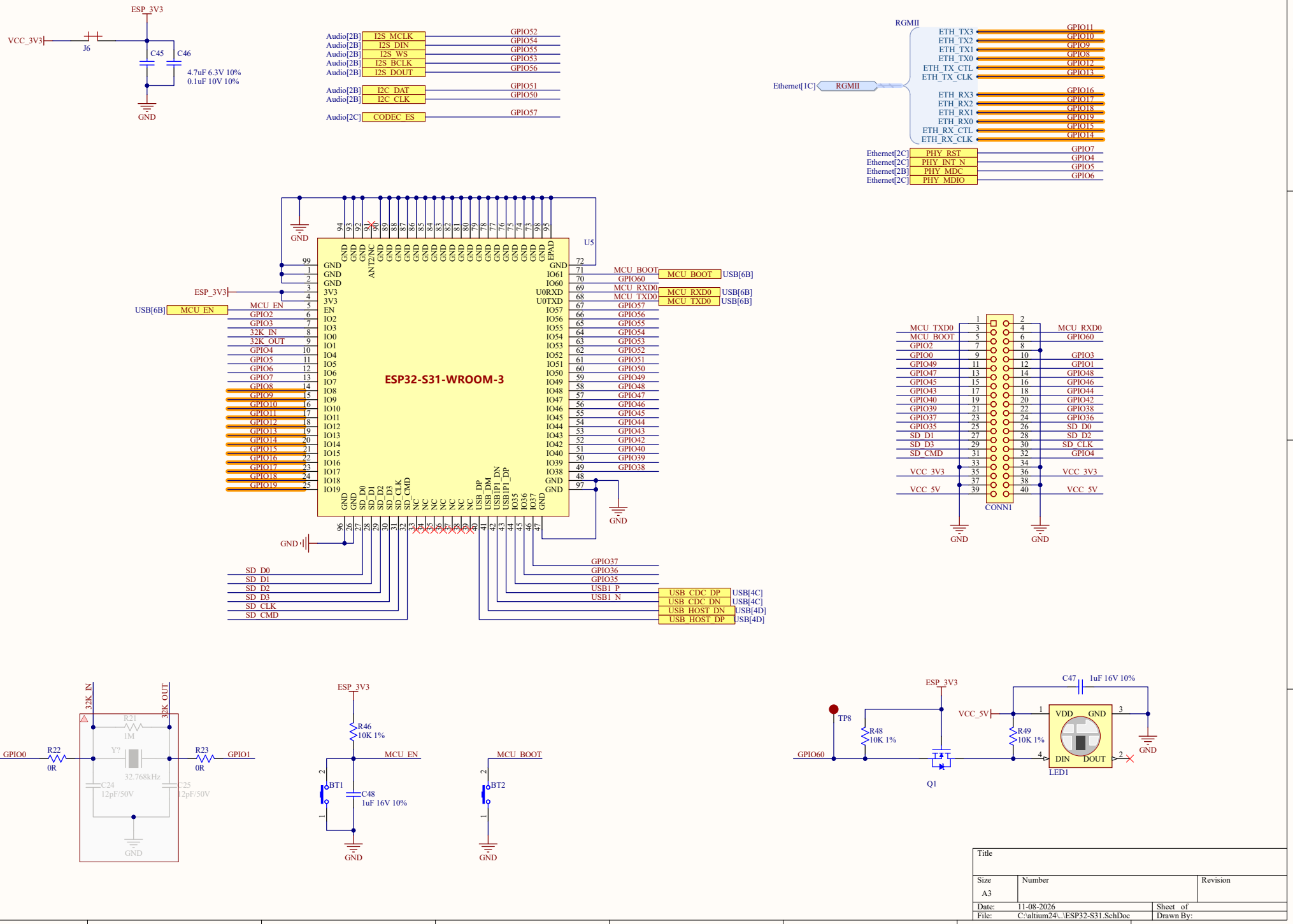
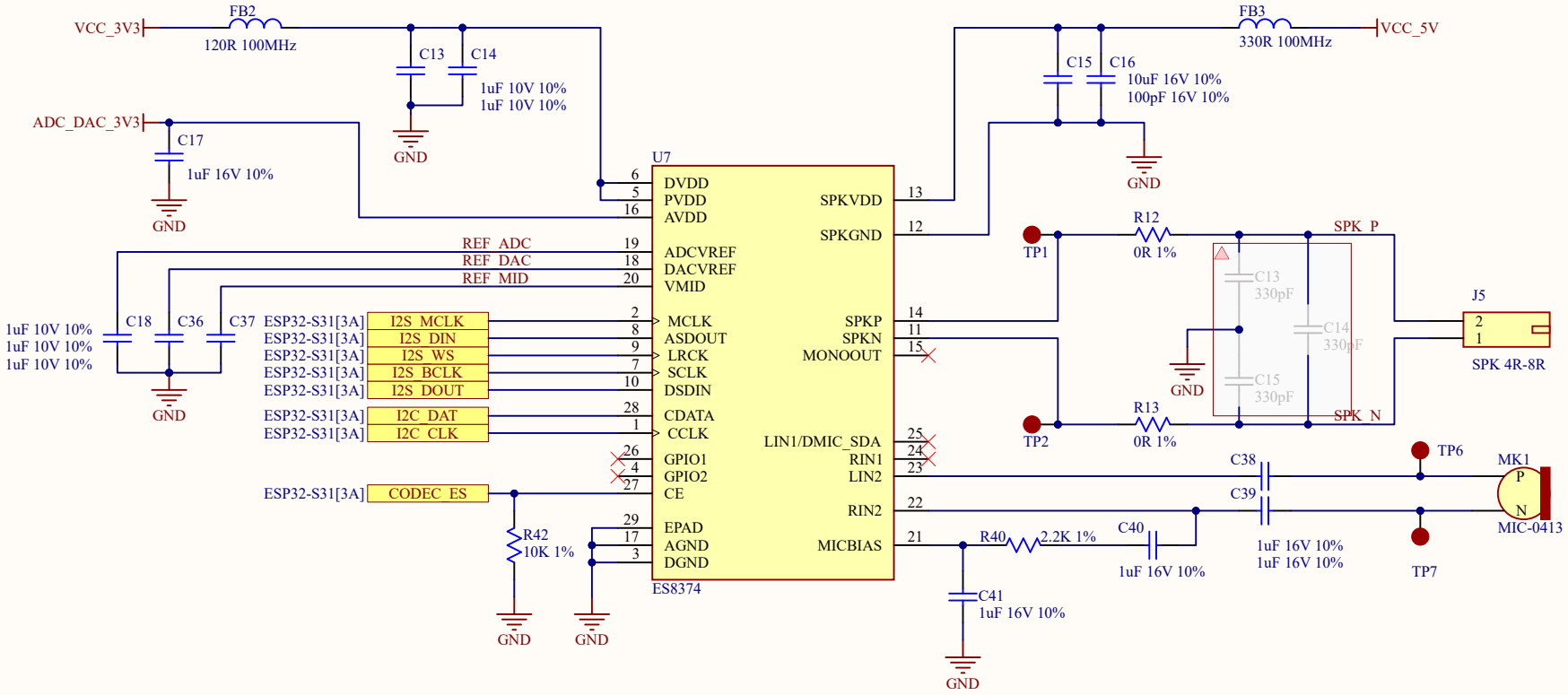
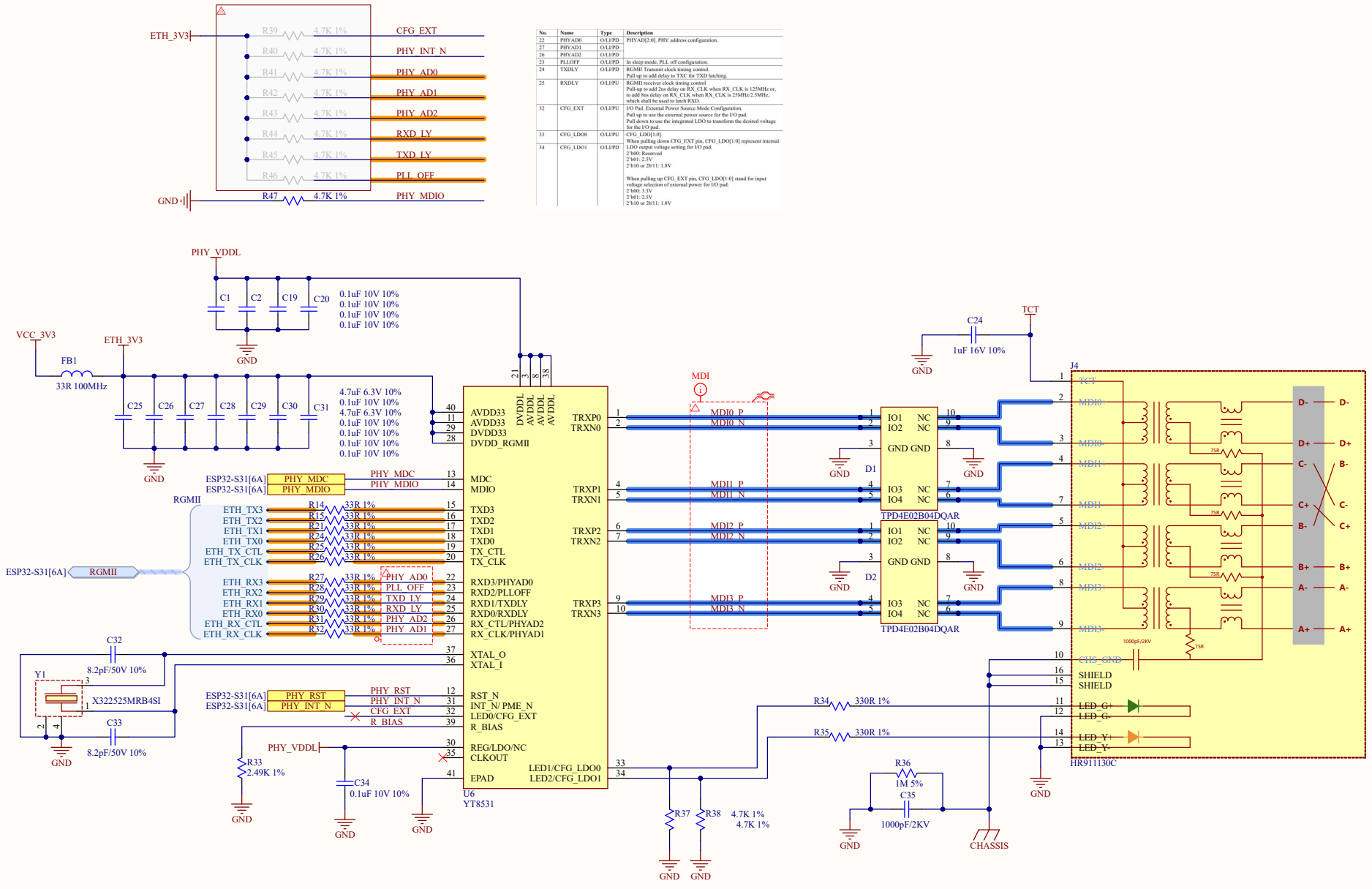

VBUS_UART
VBUS_CDC
IC1
VIN1
VIN2
VOUT1
VOUT2
ON
GND
GND2
ST
LM66200DRLR
VCC_5V
TP4
C3
0.1uF 16V 10%
C4
10uF 10V 20%
R1 10K 1%
U1
VIN
EN
SW
GND
FB
TLV62569PDDCR
L1
2.2uH 20%
VCC_3V3
TP5
R3
100K 1%
R4
22.1K 1%
C7
22pF
C5
C6
C7
C8
2.2uF 16V 10%
2.2uF 16V 10%
2.2uF 16V 10%
2.2uF 16V 10%
R2
6.8K 1%
LD1
R5 1K 1%
VDD
U2
RST
VREGIN
DCD
RI/CLK
D+
D-
VBUS
GPIO.0/TXT
GPIO.1/RXT
GPIO.2/RS485
GPIO.3/WAKEUP
GPIO.4
GPIO.5
GPIO.6
NC
SUSPEND
SUSPEND
CHREN
CHR0
CHR1
TXD
RXD
RTS
CTS
DSR
DTR
CP2102N-A02-GQFN28
C9
0.1uF 16V 10%
C10
4.7uF 16V 10%
C11
2.2uF 10V 10%
USB0_P
USB0_N
R6
499R 1%
MCU_RXD0
MCU_TXD0
ESP32-S31[5B]
RTS
DTR
R7
10K 1%
MCU_EN
MCU_BOOT
ESP32-S31[2B]
BC847
U3
R10
10K 1%
J1
VBUS
DP
DN
CC1
CC2
SBU1
SBU2
SHIELD
D3
D4
D5
LESD5D5.0CT1G
R8
5.1K 1%
R9
5.1K 1%
R43
22.1K 1%
R44
C44
2.2uF 10V 10%
47.5K 1%
VBUS_CDC
C12
0.1uF 16V 10%
R11
33R
R16
33R
USB1_P
USB1_N
USB_CDC_DP
USB_CDC_DN
ESP32-S31[5C]
D6
D7
D8
R17
5.1K 1%
R18
5.1K 1%
J2
TP3
R19
LIMIT 500mA
10K 1%
VCC_5V
U4
OUT
IN
FLT
EN/EN
GND
TPS2051CDBVR
R20
10K 1%
R39
10K 1%
C21
10uF 10V 20%
VBUS_HOST
J3
USB-A1SSW6
VCC
D-
D+
GND
USB_HOST_P
USB_HOST_N
USB_HOST_DP
USB_HOST_DN
C22
C23
22uF 10V 20%
0.1uF/16V 10%
D9
D10
D11
Title
Size
A3
Number
Revision
Date:
11-08-2026
Sheet of
File:
C:\altium24\..\USB.SchDoc
Drawn By:



Title		
Size A3	Number	Revision
Date: 11-08-2026		Sheet of
File: C:\altium24\..\ESP32-S31.SchDoc		Drawn By:



Title		
Size: A4	Number	Revision
Date: 11-08-2026		Sheet of
File: C:\altium24\..\Audio.SchDoc		Drawn By:



No.	Name	Type	Description
22	PHYAD0	O/LI/PD	PHYAD[2:0]. PHY address configuration.
27	PHYAD1	O/LI/PD	
26	PHYAD2	O/LI/PD	
23	PLLOFF	O/LI/PD	In sleep mode, PLL off configuration.
24	TXDLY	O/LI/PD	RGMII Transmit clock timing control. Pull up to add delay to TXC for TXD latching.
25	RXDLY	O/LI/PU	RGMII receiver clock timing control Pull-up to add 2ns delay on RX_CLK when RX_CLK is 125MHz or, to add 8ns delay on RX_CLK when RX_CLK is 25MHz/2.5MHz, which shall be used to latch RXD.
32	CFG_EXT	O/LI/PU	I/O Pad. External Power Source Mode Configuration. Pull up to use the external power source for the I/O pad. Pull down to use the integrated LDO to transform the desired voltage for the I/O pad.
33	CFG_LDO0	O/LI/PU	CFG_LDO[1:0]. When pulling down CFG_EXT pin, CFG_LDO[1:0] represent internal LDO output voltage setting for I/O pad: 2'b00: Reserved 2'b01: 2.5V 2'b10 or 2b'11: 1.8V When pulling up CFG_EXT pin, CFG_LDO[1:0] stand for input voltage selection of external power for I/O pad: 2'b00: 3.3V 2'b01: 2.5V 2'b10 or 2b'11: 1.8V
34	CFG_LDO1	O/LI/PD	

Title			
Size A3	Number		Revision
Date: 11-08-2026		Sheet of	
File: C:\altium24\..\Ethernet.SchDoc		Drawn By:	